

Organic Chemistry

19Q24119

Southeast University

School of Chemistry and Chemical Engineering

Update: March 8, 2026



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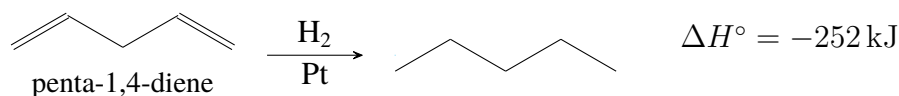
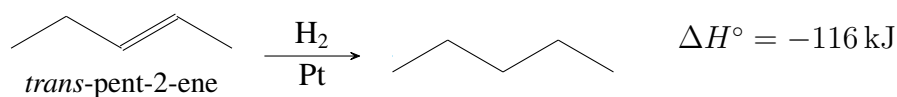
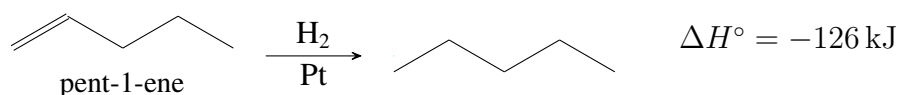
15 Conjugated Systems, Orbital Symmetry, and Ultraviolet Spectroscopy

15.1 Introduction: Conjugated Systems 20260303

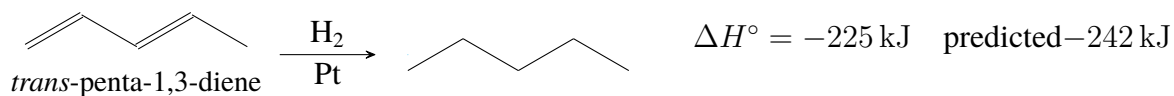
1. Conjugated double bonds: separated by one single bond.
2. Isolated double bonds: separated by two or more single bonds.
3. Cumulated double bonds: allenes, adjacent to each other.
4. Conjugated double bonds are more stable than isolated double bonds.

15.2 Stabilities of Dienes 20260303

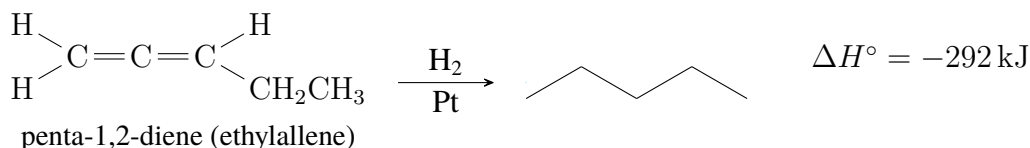
1. Using heats of hydrogenation to compare the relative stabilities of alkenes. The more stable the compound, the less exothermic during hydrogenation.



2. For conjugated dienes, the heat of hydrogenation is less than the sum for the individual double bonds, showing that the conjugated diene has extra stability.



3. The heat of hydrogenation of penta-1,2-diene is larger than any of the other pentadienes.



4. (1) The cumulated double bonds are less stable than isolated double bonds and much less stable than conjugated double bonds.
(2) Corresponding alkynes are less stable than alkenes.
(3) More substitution on double bonds or triple bonds increases the stability.

Example 15.1 Rank each group of compounds in order of increasing heat of hydrogenation.

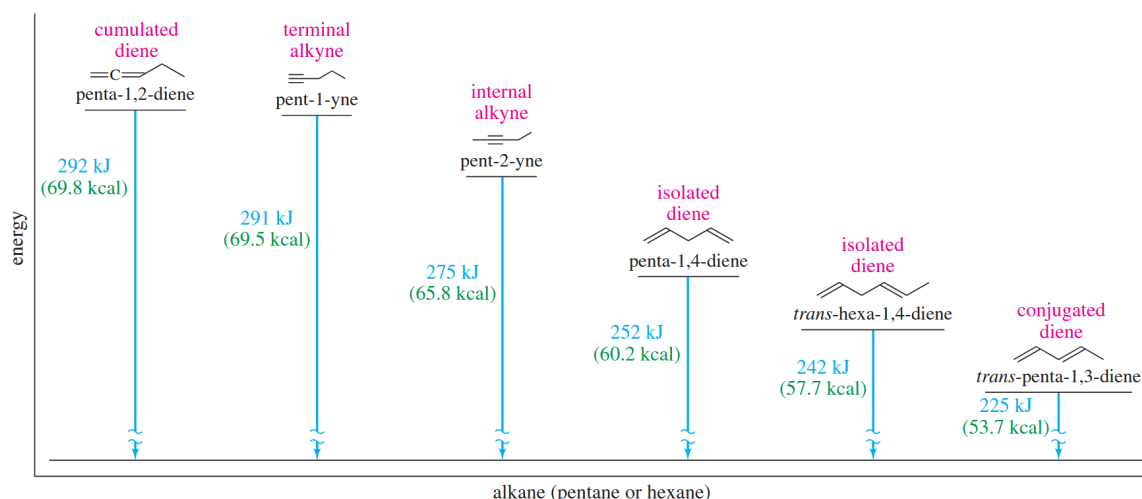


Figure 15.1: Relative Stabilities of Dienes and Alkynes

hexa-1,2-diene; hexa-1,3,5-triene; hexa-1,3-diene; hexa-1,4-diene; hexa-1,5-diene; hexa-2,4-diene

Solution: to be filled with chemfig

Example 15.2 In a strongly acidic solution, cyclohexa-1,4-diene tautomerizes to cyclohexa-1,3-diene. Propose a mechanism for this rearrangement, and explain why it is energetically favorable.

Solution: to be filled with chemfig

15.3 Molecular Orbital Picture of a Conjugated System 20260303 20260305

The extra stability of conjugated systems can be explained by examining their molecular orbitals.

15.3.1 Structure and Bonding of Buta-1,3-diene

1. The 1.48 Å central carbon–carbon single bond is shorter than the 1.54 Å bonds typical of alkanes.
2. Electrons are delocalized over molecules, creating some pi overlap and pi bonding in the bond.

15.3.2 Constructing the Molecular Orbitals of Buta-1,3-diene

1. Molecular orbitals(MO)

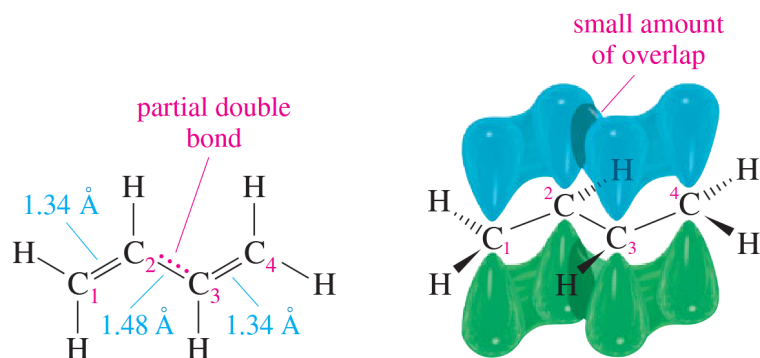


Figure 15.2: Structure of Buta-1,3-diene

2. bonding, nonbonding, and antibonding orbitals
3. energies symmetrically distributed
4. energy of isolated p orbital in the middle of the MO energy levels
5. The bonding MOs are filled and the antibonding MOs are vacant.
6. The average energy of the electrons is slightly lower in butadiene. This lower energy is the resonance stabilization of the conjugated diene.
7. Conformations of 1,3-Butadiene: *s*-trans and *s*-cis, the former, less crowded, is more stable than the latter.
8. The *s*-cis and *s*-trans conformers of butadiene (and all the skew conformations in between) easily interconvert at room temperature (The basis of DA reaction).

15.4 Allylic Cations 20260305

1. Conjugated compounds undergo a variety of reactions, many of which involve intermediates that retain some of the resonance stabilization of the conjugated system.
2. allyl group: $\text{CH}_2=\text{CH}-\text{CH}_2-$
3. Allylic cations and radicals are stabilized by delocalization.
4. Double bond interact with cations(exchange).
5. Stability of carbocations: $\text{H}_3\text{C}^+ < 1^\circ < 2^\circ$, allyl $< 3^\circ$, substituted allylic

15.5 1,2- and 1,4-Addition to Conjugated Dienes 20260305

1. Electrophilic addition to the double bond produces the most stable intermediate.
2. For conjugated dienes, the intermediate carbocation is a resonance stabilized allylic cation.
3. MECHANISM 15-1 1,2- and 1,4-Addition to a Conjugated Diene
 - (1) Protonation of one of the double bonds forms a resonance-stabilized allylic cation.

- (2) A nucleophile attacks at either electrophilic carbon atom.

15.6 Kinetic versus Thermodynamic Control in the Addition of HBr to Buta-1,3-diene 20260305

1. $-80\text{ }^{\circ}\text{C}$ 1,2-addition (kinetic control), $40\text{ }^{\circ}\text{C}$ 1,4-addition (thermodynamic control)
2. The 1,2-addition will be the faster addition at any temperature (lower E_a , secondary carbocation intermediate), but the 1,4-addition will be more stable (higher E_a , but more stable, more substituted alkene product).
3. The first step of the reaction is kinetically controlled, the second step is thermodynamically controlled.
4. Thermodynamic control at $40\text{ }^{\circ}\text{C}$
5. Kinetic control at $-80\text{ }^{\circ}\text{C}$

Example 15.3 hint: In many cases, an allylic carbocation is more stable than a bromonium ion.

15.7 Allylic Radicals 20260305

1. Allylic radicals are stabilized by resonance delocalization.
2. In a free-radical bromination of cyclohexene, substitution occurs entirely at the allylic position.
3. MECHANISM 15-2 Free-Radical Allylic Bromination
 - (1) Initiation: Formation of radicals.

Note Cleavage should happen in nonpolar solvents. e.g. CCl_4

4. Stability of free radicals: $\text{CH}_3 < 1^{\circ} < 2^{\circ} < 3^{\circ} < \text{allylic} < \text{benzylic}$
5. Bromination Using N-Bromosuccinimide (NBS)

Example 15.4 prob 15-11

15.8 Molecular Orbitals of the Allylic System 20260305

15.9 $\text{S}_\text{N}2$ Displacement Reactions of Allylic Halides and Tosylates 20260305

1. Allyl bromide reacts with nucleophiles by the mechanism about 40 times faster than n-propyl bromide.
2. Allylic halides are so reactive that they couple with Grignard and organolithium reagents